

VALIDATION AND JUSTIFICATION OF DESIGN NUMBERS

MastitisGuard AI — Why Each Number and Choice Was Made

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Problem Statement Number: 26109

Journey stage: Verified with the market

1. Introduction

Every dataset size, threshold, weighting, and architectural choice in this project has a specific reason behind it. This document exists so that any number quoted elsewhere in the project's materials — the dataset size, the AUC, the class weights, the decision threshold — can be traced back to why that particular value was chosen, rather than presented as an unexplained result. Every figure below is quoted identically in the ML Model Documentation and the Decision-Making and PoC document.

2. Purpose of This Document

To pre-empt the most natural follow-up question a judge or reviewer can ask about any quantitative claim in this project: "why this number, and not another one?" Each row below states the number or choice, and the specific reasoning or calculation behind it.

3. Design Decisions and Their Justification

Number / Choice	Why This Value Was Chosen
8,000 readings (400 cows × 20 days)	Enough rows for CatBoost to learn stable patterns across 24 features (21 numeric + 3 categorical) without overfitting a small sample; 400 independent cows gives a meaningful train/test split by animal (300 train / 100 test) rather than too few animals to generalize across.
~30% of cows develop mastitis in the simulated window	Deliberately higher than real-world incidence so the small number of true positive events is large enough to train on at all; the resulting ~8% positive-reading rate after windowing still reflects realistic sparsity, and is compensated for by class weighting rather than by inflating incidence further.
1–5 day pre-clinical drift window (demonstrated); 7–14 days (literature target)	The 1–5 day window is the window actually present in the synthetic dataset and is the figure reported as a result throughout the ML Model Documentation. The 7–14 day figure, cited in the Technical Dossier, is the longer-range target drawn from veterinary literature that Phase 1 real-farm data collection is designed to work toward — the two numbers are never used interchangeably in this project's documents.
Class weights [1, 12]	Derived directly from the measured positive rate in the training data (~8%): the inverse class frequency is $1/0.08 \approx 12.5$, rounded to 12. This is a principled choice based on the actual data distribution, not an arbitrary tuning value.
Decision threshold 0.729	Not chosen in advance — selected by sweeping all possible thresholds along the precision-recall curve on validation data and picking the value that maximizes F1 score. This is a data-derived value, recomputed automatically any time the model is retrained.

Number / Choice	Why This Value Was Chosen
Train/test split by Cow ID, not by row	Splitting by row would let the model see other days from the same cow in both training and test data, allowing it to partly memorize that cow's baseline rather than learn a generalizable pattern. Splitting by animal guarantees the test cows are entirely unseen during training, which is the only way to honestly measure how the model performs on a new animal.
Fusion weight swept from 0.0 to 1.0 in steps of 0.05	A step size of 0.05 gives 21 candidate weights — fine enough granularity to find a stable optimum without the computational cost of a continuous search, which is unnecessary given the fusion score is a simple weighted average of two bounded [0,1] values.
CatBoost chosen over XGBoost / Random Forest / a neural network	CatBoost natively handles the categorical features (breed, lactation stage, season) without manual encoding, and — critically — has a built-in exporter to standalone C++, which is what makes offline ESP32 deployment possible without a separate conversion pipeline such as TensorFlow Lite would require. A neural network was not used because tabular data of this size and structure is well established in the literature to favor gradient-boosted trees over deep learning, and trees are far cheaper to export for microcontroller inference.
ROC-AUC and F1 chosen as primary metrics	Plain accuracy is misleading on an imbalanced dataset (a model predicting "healthy" for every reading would already be ~92% "accurate" while useless). ROC-AUC measures ranking quality independent of any single threshold, and F1 balances precision and recall at the threshold actually used in deployment — together they give an honest picture that accuracy alone would hide, and neither is quoted as a real-world field-accuracy figure (see ML Model Documentation, Section 7).
₹2,510 / ₹2,110 / ₹950 per-unit costs	Each figure is a direct sum of the itemized components in the Bill of Materials (Sections 2–4); no rounding or estimation is applied beyond the individual component prices quoted by Indian domestic electronics distributors.
Three physical sensing units (Wearable, Milk Unit, Shed ESP) rather than one combined device	Matches where the physical measurement actually happens: behavioral signals can only be captured on the animal itself, milk-quality signals can only be captured at the point of milking , and the two happen at different times and locations during a farm's daily routine. A single combined device would require either the animal to be moved to a fixed processing station continuously, or the processing unit to be worn — neither of which is practical. This also directly follows the multi-sensor-fusion approach recommended in the cited literature, which identifies single-parameter sensors (e.g. handheld EC meters) as producing high false-positive rates when used alone.

4. What Is Still an Assumption, Stated Plainly

In the interest of the same honesty this document asks for elsewhere, the following things in the current prototype are reasoned estimates rather than empirically measured facts, and are flagged as such wherever they appear in other project documents:

- The physiological healthy ranges and mastitis-related shifts used to generate the training dataset are drawn from the cited literature, but the literature covers **general bovine physiology rather than this exact sensor combination** — so the synthetic dataset is grounded, not verified against real sensor readings from real cows.

- The milk **electrical conductivity (EC)** feature in the model schema (ML Model Documentation, Section 4) has no corresponding physical sensor in the current Bill of Materials — it is retained for a planned Phase 2 EC probe addition and should not be read as a **currently-sensed input**.
- The Serviceable Addressable Market figure in the Market Research document is built from an assumed minimum viable herd size (3+ animals), since no published dataset breaks down cooperative membership by herd size.
- The self-learning retraining cadence ("monthly, or once enough new confirmed cases accumulate") is a proposed operating rule, not yet validated against how quickly real farms will actually generate vet-confirmed outcomes.
- The **0.964 ROC-AUC** is a synthetic held-out-cow result. **It is never quoted in this project's documents as a real-world field accuracy figure**; that validation is the Phase 1 milestone described in the Market Research document, Section 4, and the ML Model Documentation, Section 11.

5. Conclusion

Treating every number in this project as something that must be defended, not just reported, is what separates a demonstrated result from a plausible-sounding claim. Where a number comes directly from measured data (the AUC, the decision threshold, the class weights, the per-unit costs), that is stated plainly, and the **identical figure** is used in every other document in this set. Where a number is instead a reasoned design target or an explicit assumption, that is stated just as plainly, so that nothing in this project is at risk of sounding more empirically certain than it actually is.

6. References

- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. *Advances in Neural Information Processing Systems (NeurIPS)*, 31, 6639–6649.
- scikit-learn developers. (2025). 3.1. Cross-validation: evaluating estimator performance — Group K-fold. *scikit-learn 1.9 documentation*. scikit-learn.org/stable/modules/cross_validation.html.